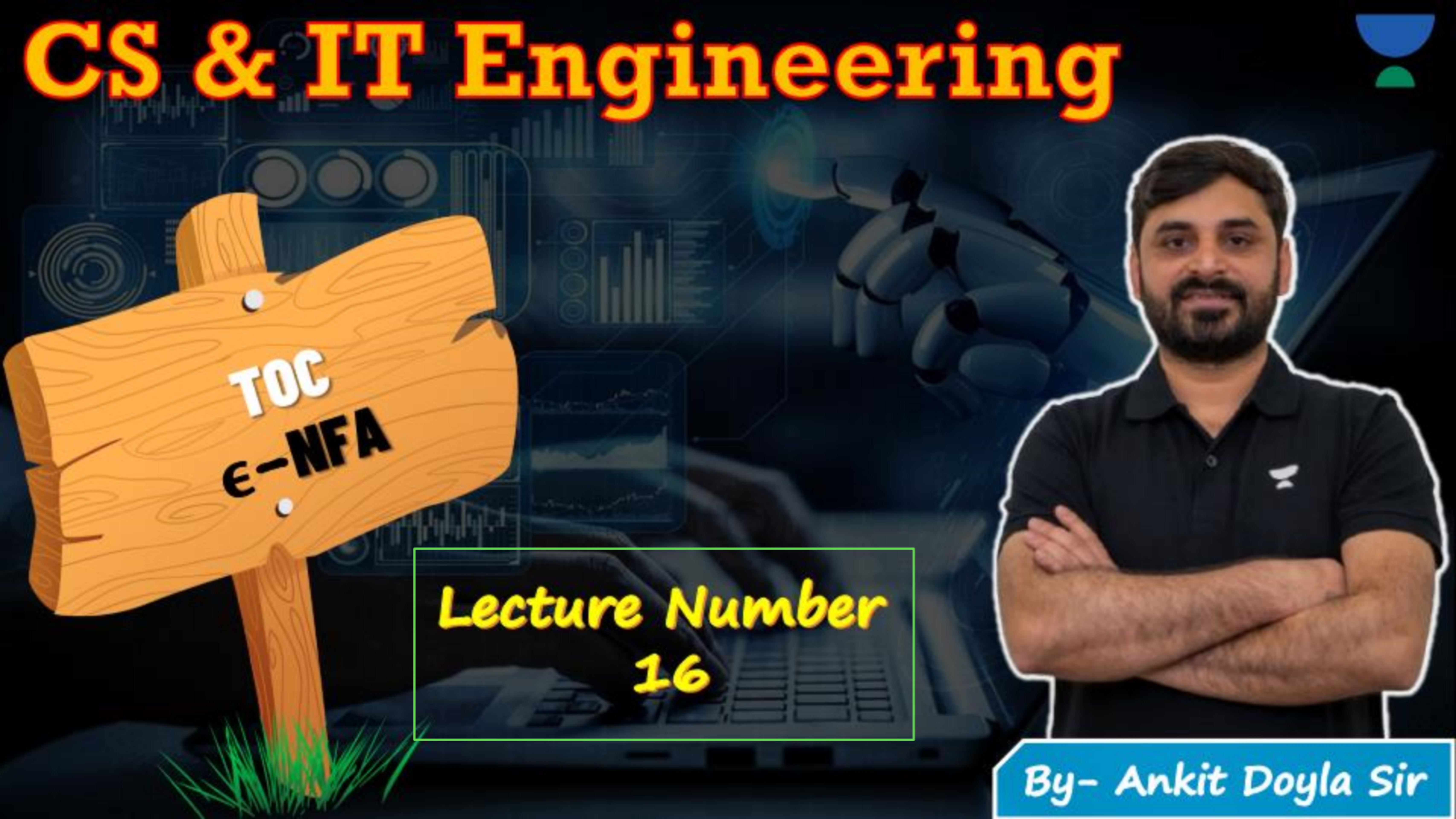
CS & IT Engineering
TOC
ϵ-NFA
Lecture Number
16
By- Ankit Doyla Sir

Topics to be Covered

1. ϵ -NFA
2. ϵ -NFA to NFA
3. ϵ -NFA to DFA

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ϵ - NFA

The NFA which has a transition even for empty string ϵ is called as ϵ-NFA.

OR

ϵ - NFA is a collection of 5 tuple

i.e. $M = \{Q, \Sigma, \delta, q_0, F\}$

$Q \longrightarrow$ set of All states

$\Sigma \longrightarrow$ input Alphabet

$q_0 \longrightarrow$ initial state

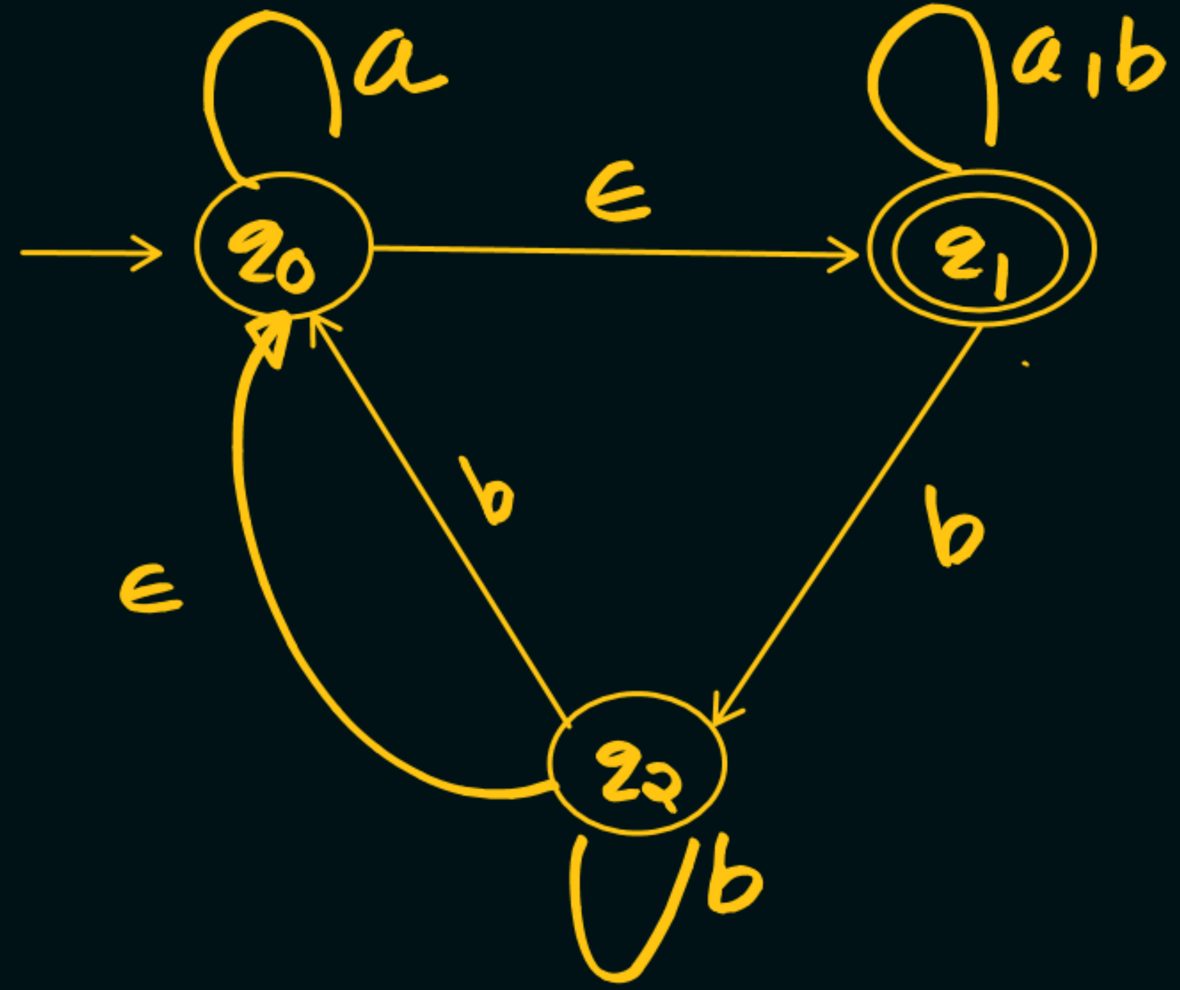
$F \longrightarrow$ set of Final state

$\delta : Q \times \Sigma \cup (\epsilon) \longrightarrow 2^Q$ is a transistion Function

$\delta : Q \times \Sigma \longrightarrow Q$ (DFA)

$\delta : Q \times \Sigma \longrightarrow 2^Q$ (NFA)

$\delta : Q \times \Sigma \cup (\epsilon) \longrightarrow 2^Q$ (ϵ NFA)



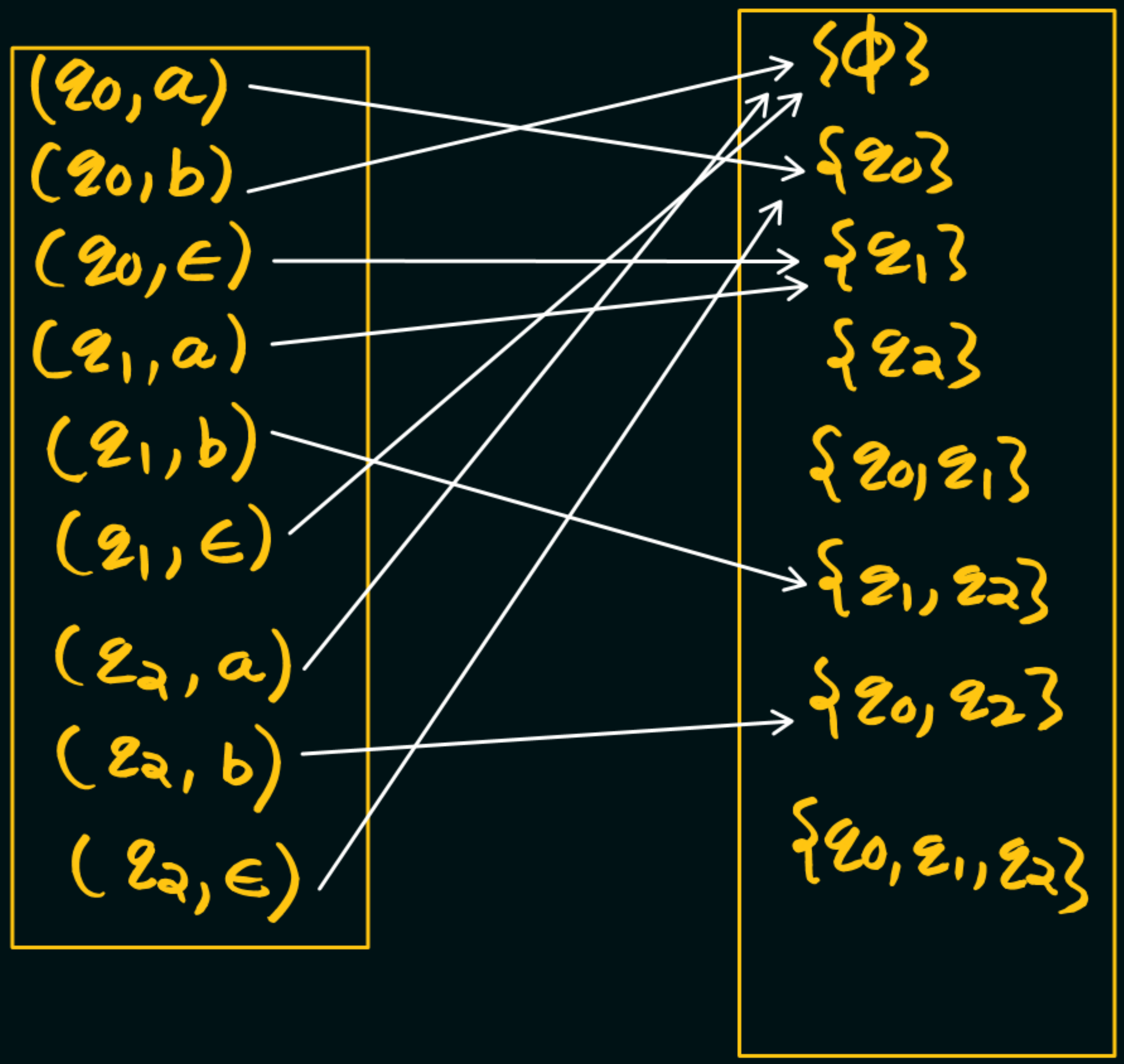
$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$q_0 = \{q_0\}$

$F = \{q_1\}$

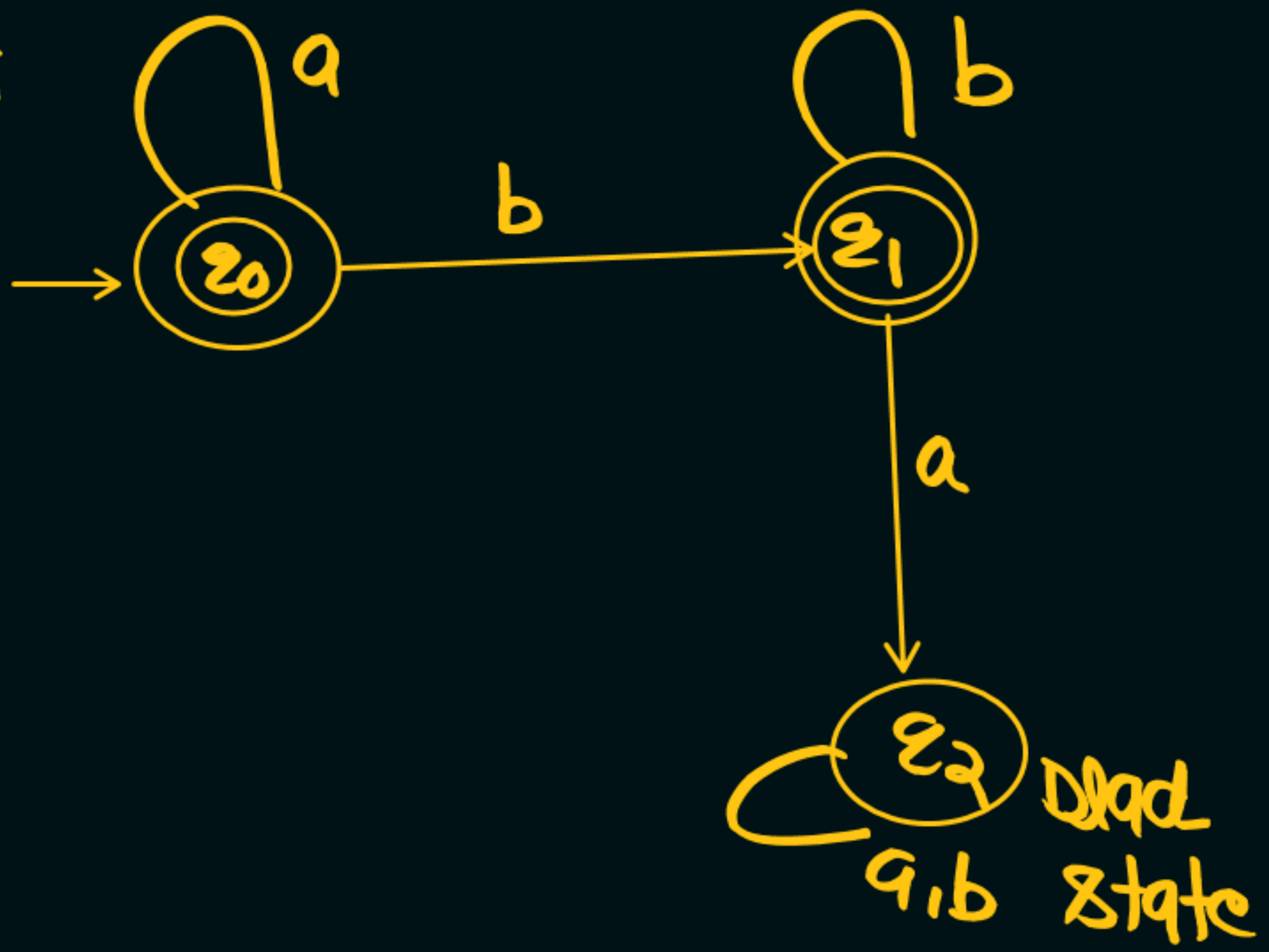
$\delta : Q \times \Sigma \cup (\epsilon) \longrightarrow 2^Q$

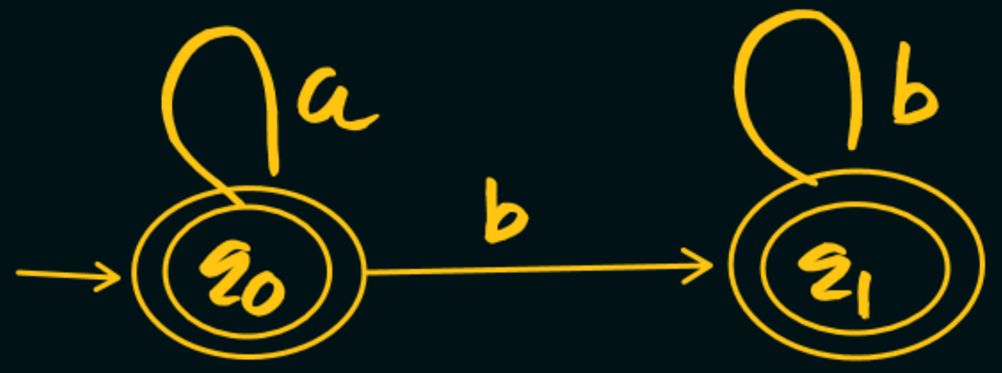


1. $L = a^m b^n \mid m \geq 0, n \geq 0$

$L = \{ \epsilon, a, aa, aaa, \ldots$
$b, bb, bbb, \ldots$
$ab, aab, abb, aabb$

DFA





NFA

$a^m \epsilon \cdot b^n = a^m b^n$

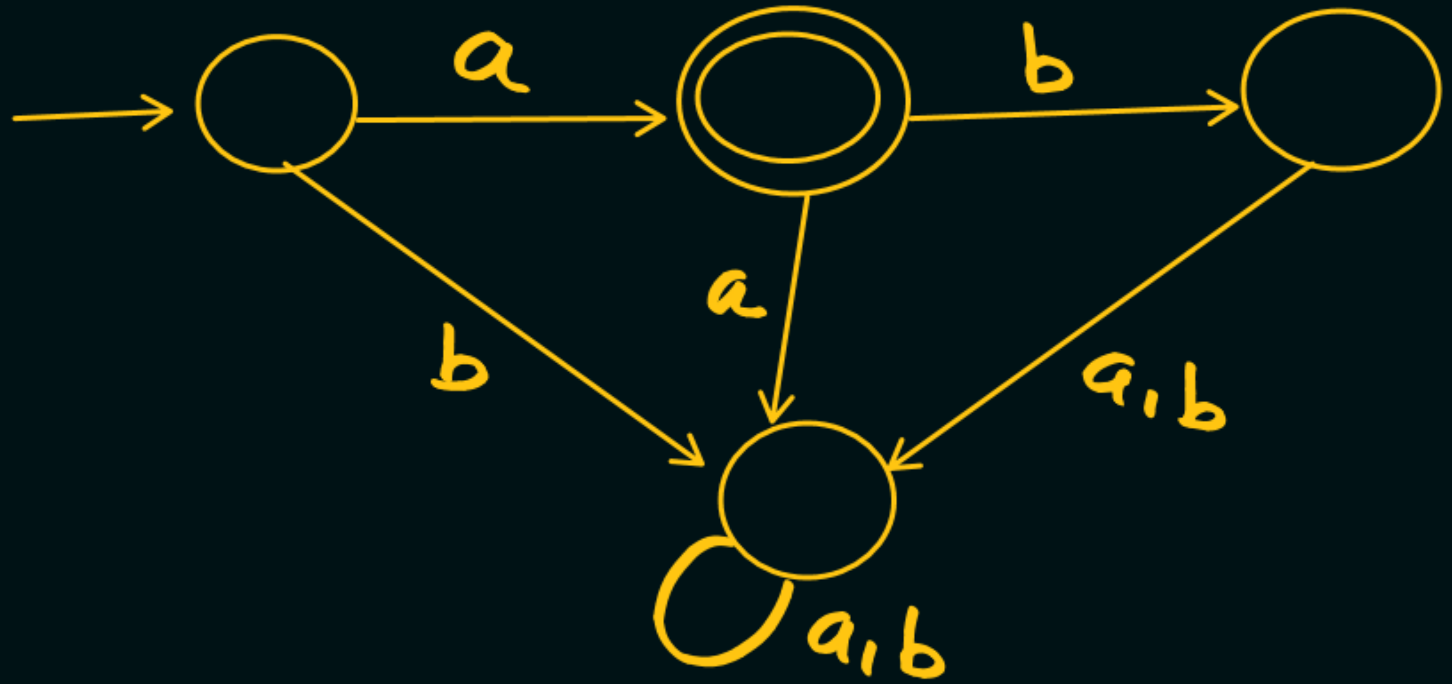
$\boxed{\text{Note:}}$ ϵ-NFA can Accept any Regular Language with only one Final state

Note

NFA can Accept any Regular Language with only one Final state

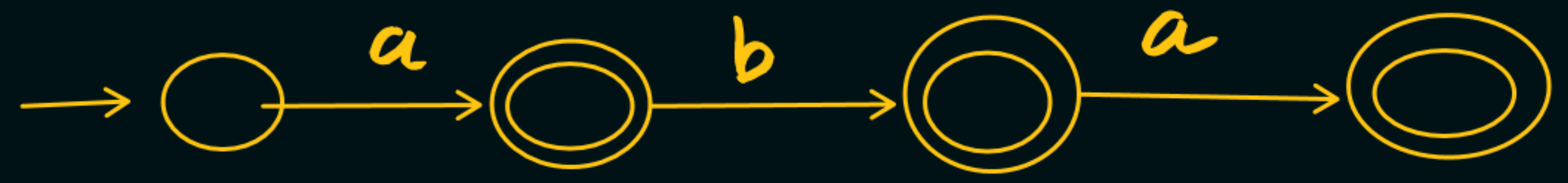
1 $L = \{a, ab\}$, $\Sigma = \{a, b\}$

DFA

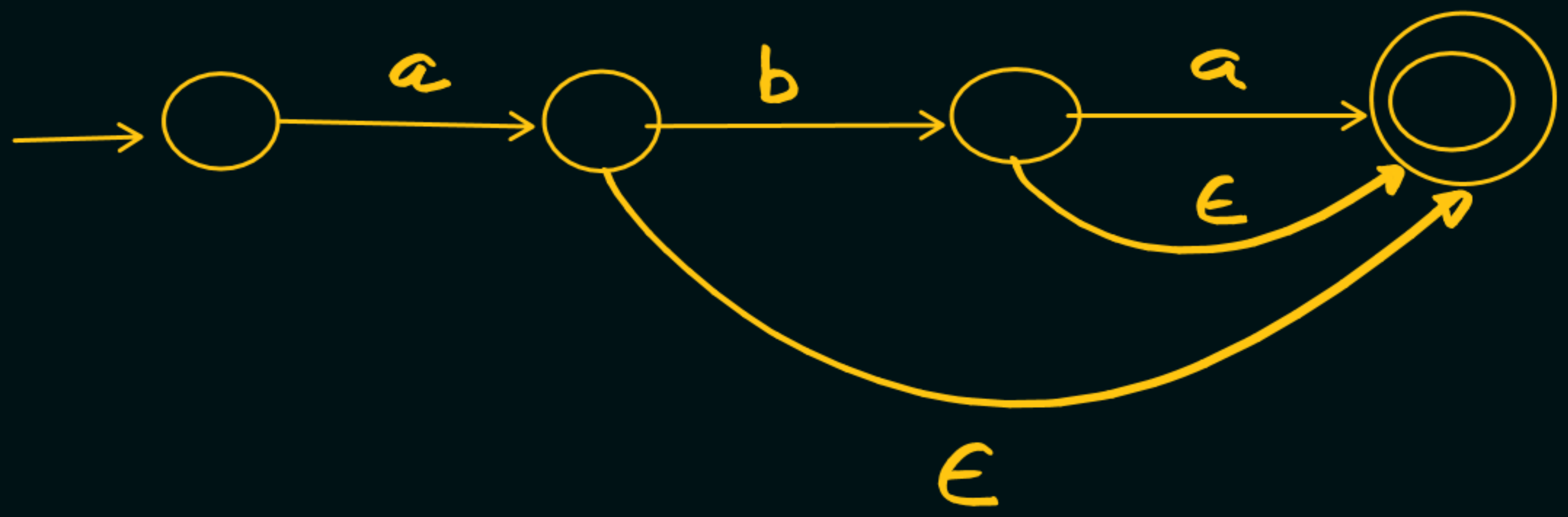


NFA

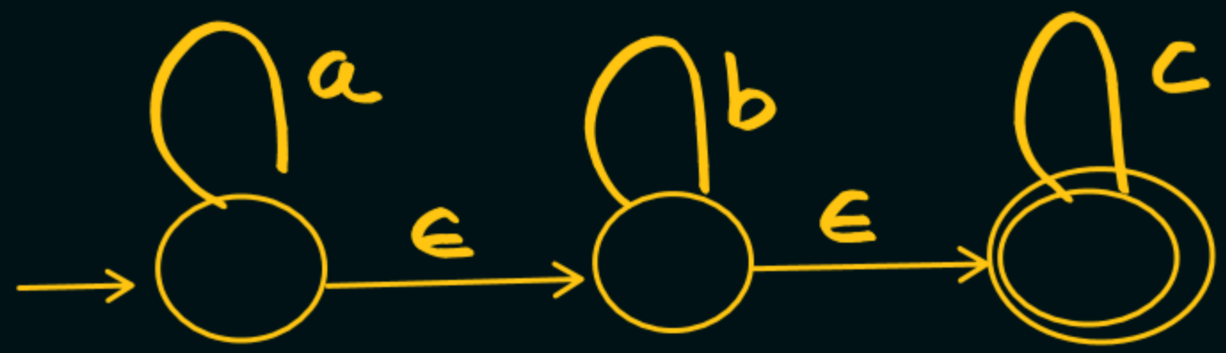
$L = \{a, ab, aba\}, \Sigma = \{a, b\}$



NFA



$L = a^m b^n c^P \mid m \geq 0, n \geq 0, P \geq 0$



ϵ-Closure of (q)

Let q is any state in ϵ-NFA, then the set of all states which are at 0 distance from state q, is called as ϵ closure.

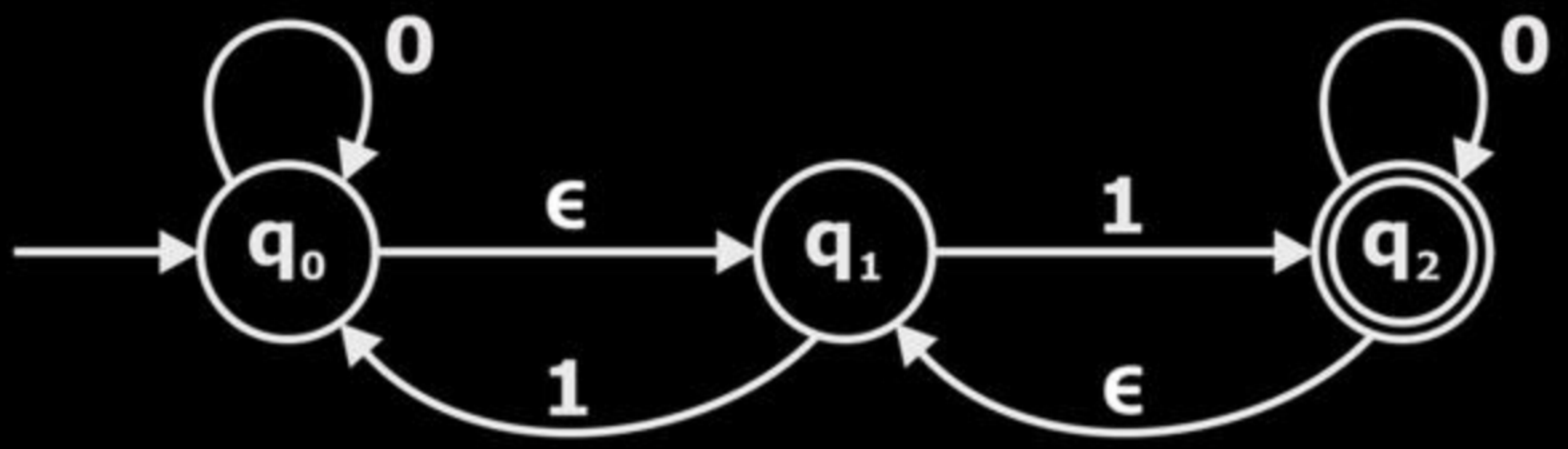
OR

Let Q is any state in NFA. The set of all states that can be reached from Q on an empty string ϵ is called as ϵ closure

Note

- every state is at 0 distance from itself.

Example-1

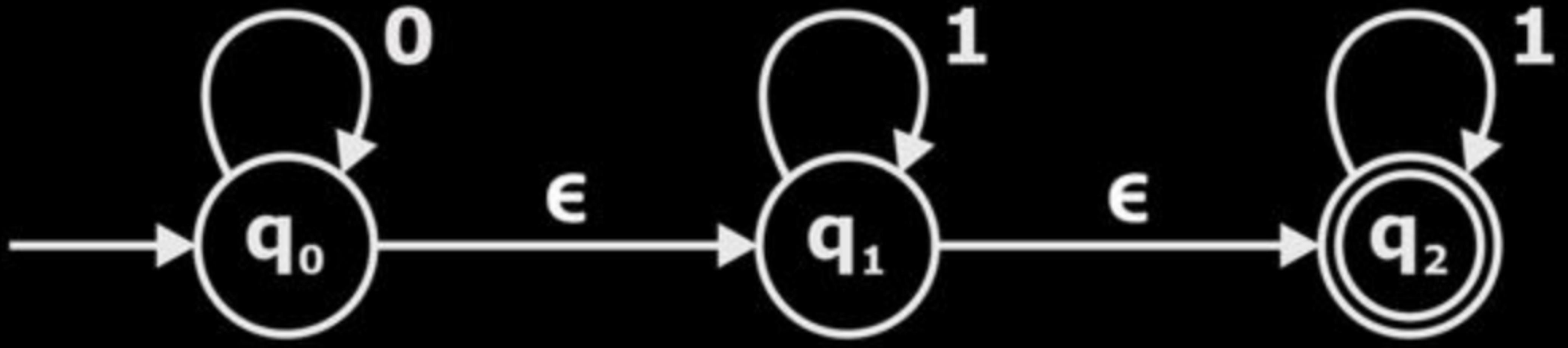


$\epsilon\ closure(q_0) = \{q_0, q_1\}$

$\epsilon\ closure(q_1) = \{q_1\}$

$\epsilon\ closure(q_2) = \{q_2, q_1\}$

Example-2

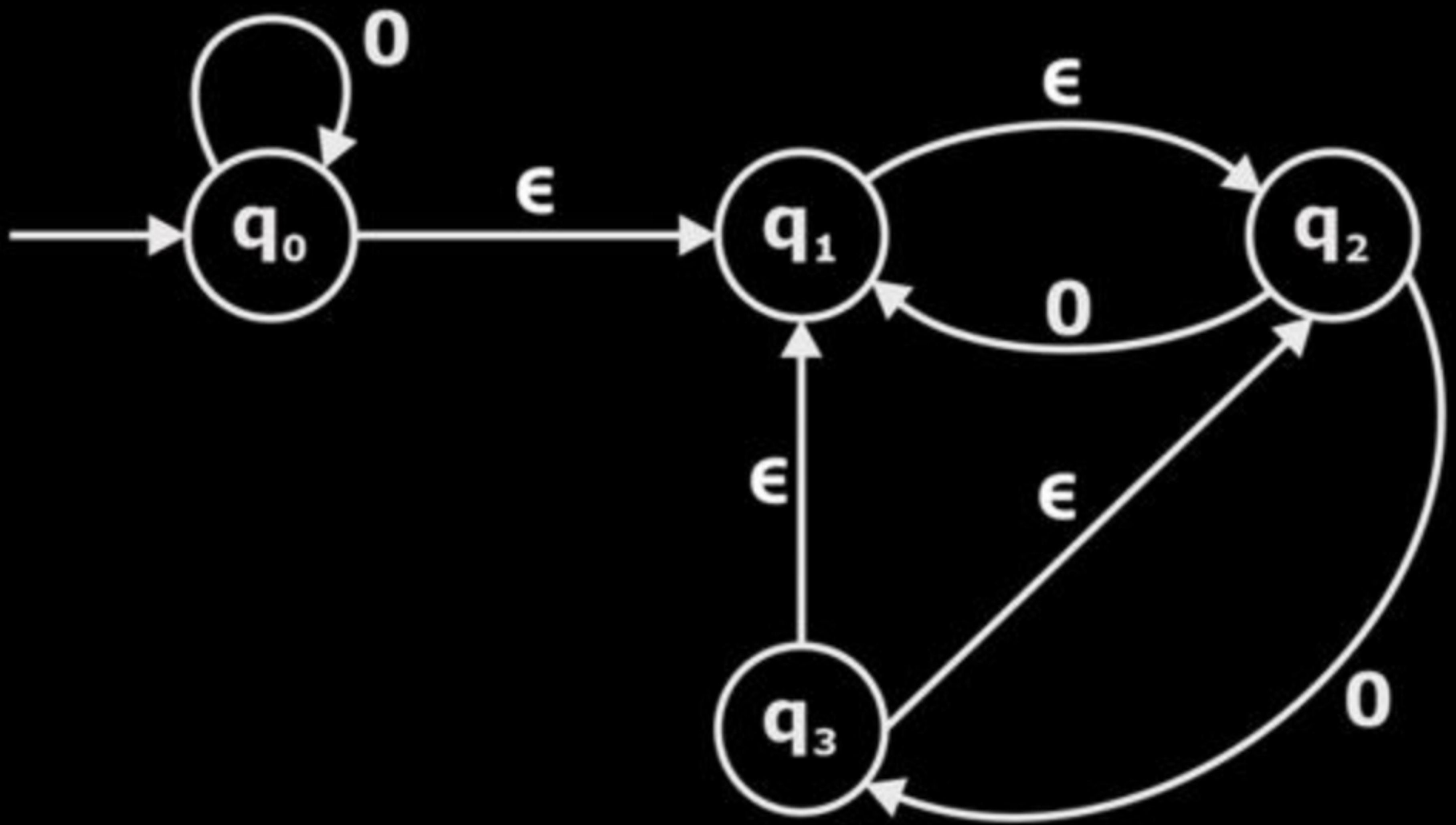


$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

Example-3



$\epsilon\ closure(q_0) = \{q_0, q_1, q_2\}$

$\epsilon\ closure(q_1) = \{q_1, q_2\}$

$\epsilon\ closure(q_2) = \{q_2\}$

$\epsilon\ closure(q_3) = \{q_3, q_1, q_2\}$

$$\begin{tikzcd} \epsilon\text{-NFA} \arrow[r] \arrow[rr, bend left] & \text{NFA} \arrow[r] & \text{DFA} \end{tikzcd}$$

$\boxed{\text{Thomson Construction}}$ $\boxed{\text{Subset Construction}}$

Conversion of ϵ-NFA to NFA

1. No change in the initial state
2. No change in the total no. of states
3. May be change in final state

Algorithm

Let $M = \{(Q, \Sigma, \delta, q_0, F) \rightarrow \epsilon\text{-NFA}$

$M' = \{(Q', \Sigma, \delta', q_0', F') \rightarrow \text{NFA}$

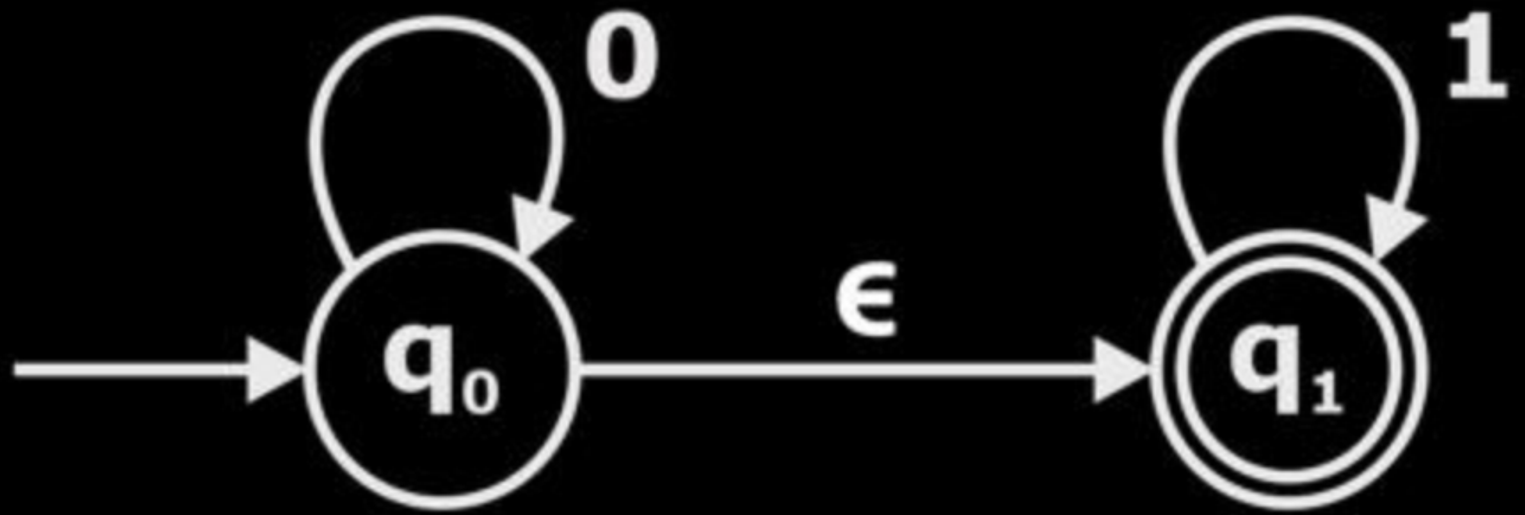
1. **Initial state**: No change in the initial state i.e.

   $\boxed{q_0' = q_0}$

2. **Construction of δ'**: $\boxed{\delta'(q, x) = \epsilon \text{ closure } \{\delta(\epsilon \text{ closure } (q), x)\}}$

3. **Final state**: Every state whose ϵ-closure contains the final state of ϵ-NFA is the final state in NFA.

Example-1



$\epsilon\text{-closure}(q_0) = \{q_0, q_1\}$

$\epsilon\text{-closure}(q_1) = \{q_1\}$

$\delta'(q,x) = \epsilon\text{closure}\{\delta(\epsilon\text{closure}(q),x)\}$

$\delta'(q_0,0) = \underbrace{\epsilon\text{closure}\{\underbrace{\delta(\underbrace{\epsilon\text{closure}(q_0)}_{1},0)}_{2}\}}_{3}$

$\epsilon\text{closure}\{\delta\ ((q_0,q_1),0\}$

$= \epsilon\text{closure}\ \{\delta(q_0,0) \cup \delta(q_1,0)$

$q_0 \cup \phi$

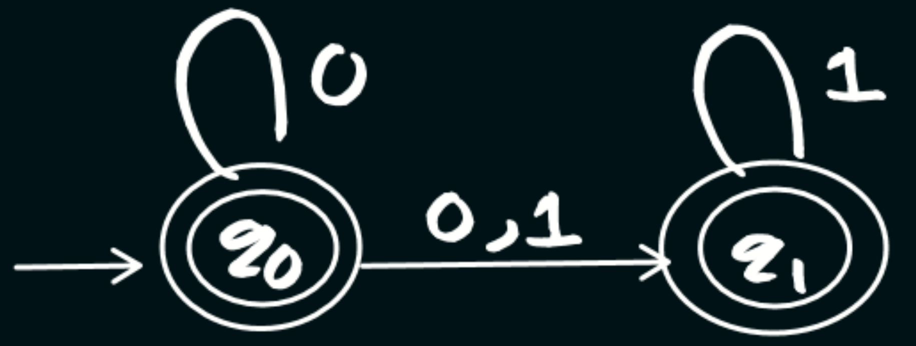
$= \epsilon\text{closure}\ (q_0)$

$\delta'(q_0,0) = \{q_0, q_1\}$

$$
\begin{tikzcd}
 & q_0 \arrow[d, "\epsilon\text{closure}"] & \\
 & (q_0, q_1) \arrow[dl, "\delta\,0"'] \arrow[dr, "\delta\,1"] & \\
q_0 \arrow[d, "\epsilon\text{closure}"] & & q_1 \arrow[d, "\epsilon\text{closure}"] \\
(q_0, q_1) & & (q_1)
\end{tikzcd}
$$

$$\boxed{\epsilon\text{closure}(\phi) = \phi}$$

$$
\begin{tikzcd}
 & q_1 \arrow[d, "\epsilon\text{closure}"] & \\
 & q_1 \arrow[dl, "\delta\,0"'] \arrow[dr, "\delta\,1"] & \\
\phi \arrow[d, "\epsilon\text{closure}"] & & q_1 \arrow[d, "\epsilon\text{closure}"] \\
\phi & & q_1
\end{tikzcd}
$$



\# states = 2

Example-1

ε-NFA

$\epsilon\text{-closure}(q_0) = \{q_0, q_1\}$

$\epsilon\text{-closure}(q_1) = \{q_1\}$

ADRULE

$0 \cdot \epsilon = 0$

$\epsilon \cdot 0 = 0$

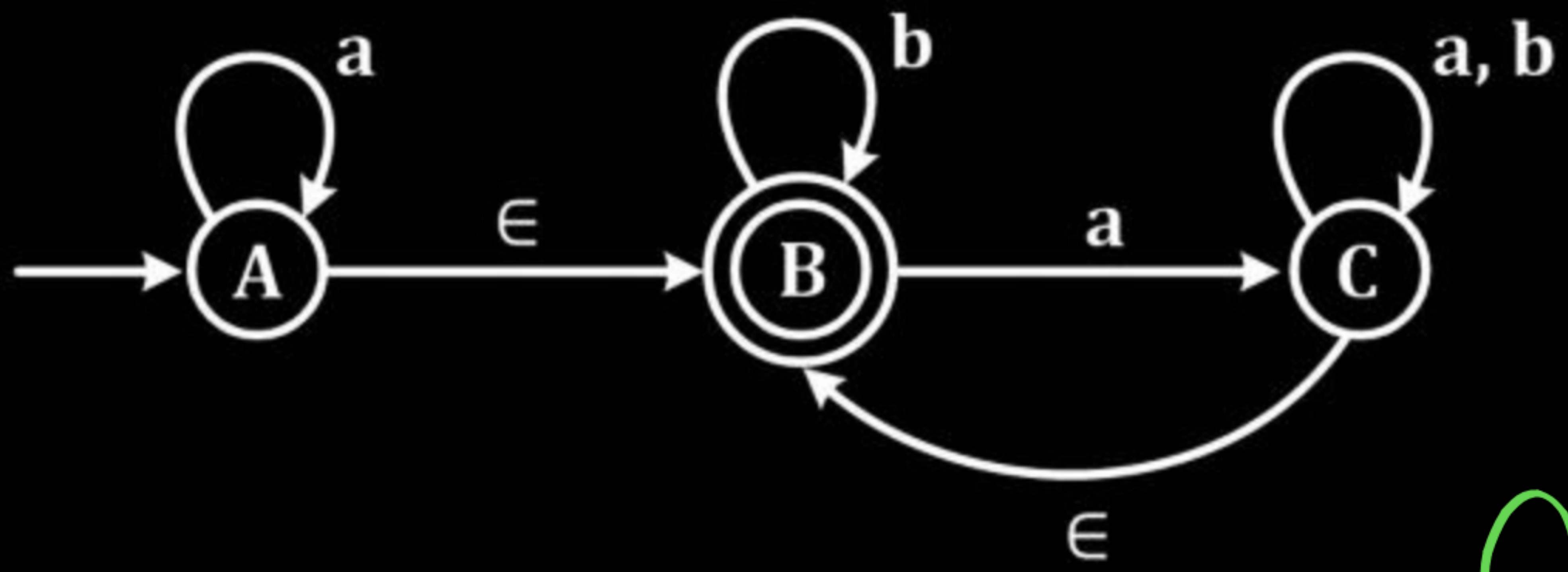
$\epsilon \cdot 0 \cdot \epsilon = 0$

$\epsilon \cdot 0 \cdot \epsilon \cdot \epsilon = 0$

NFA table

δ'	0	1
→ q_0	q_0, q_1	q_1
q_1	ϕ	q_1

Example-2



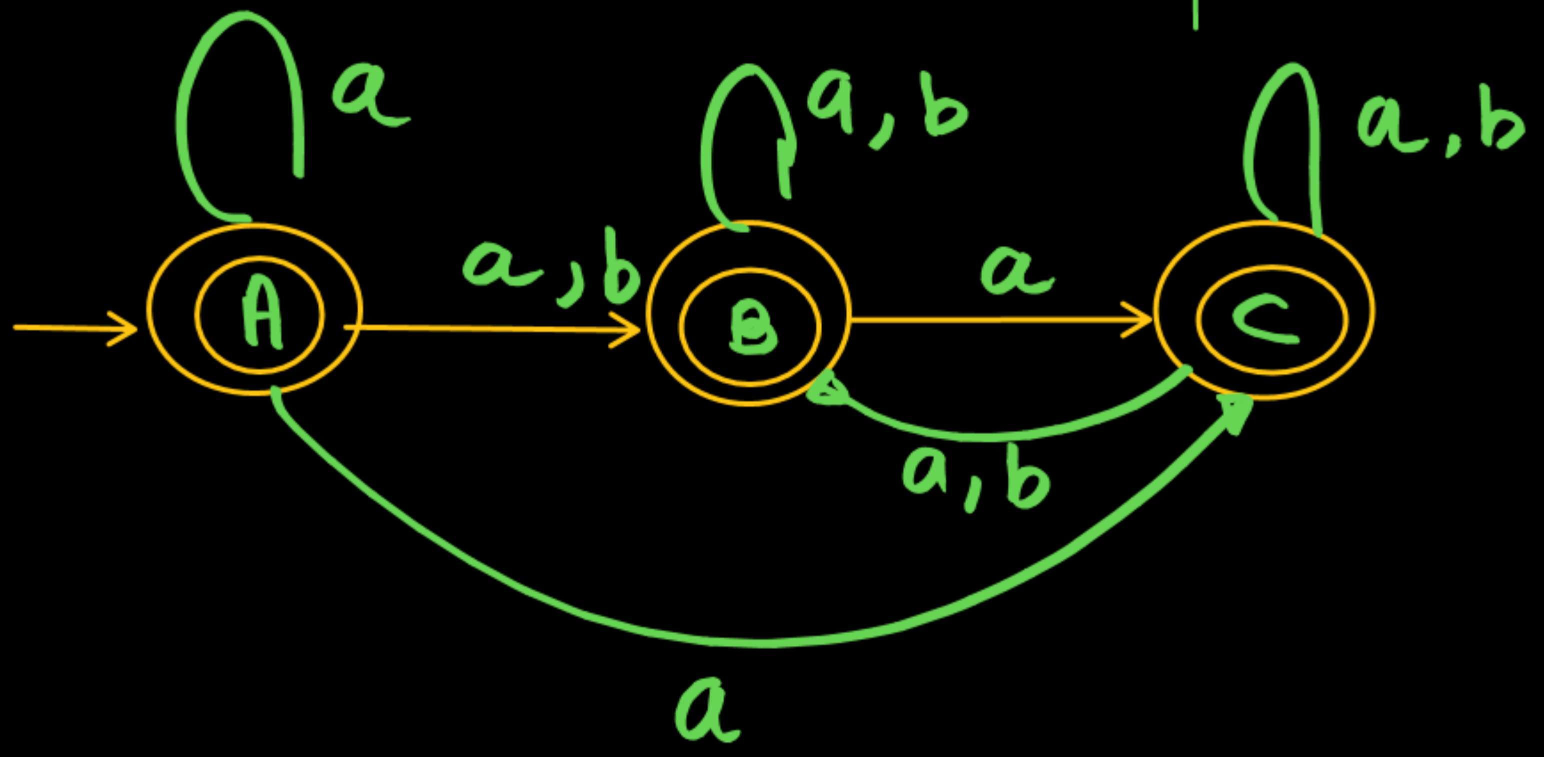
ADRule

δ'	a	b
→ A	A, B, C	B
B	C, B	B
C	C, B	C B

ϵ closure(A) = { A, B }

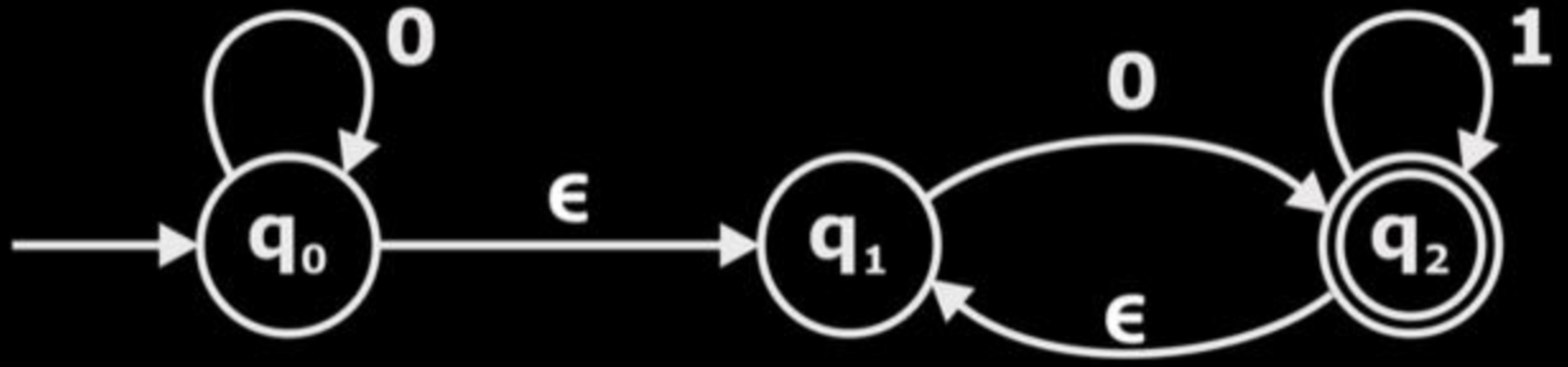
ϵ closure(B) = { B }

ϵ closure (C) = { C, B }



Example-3

ADRule



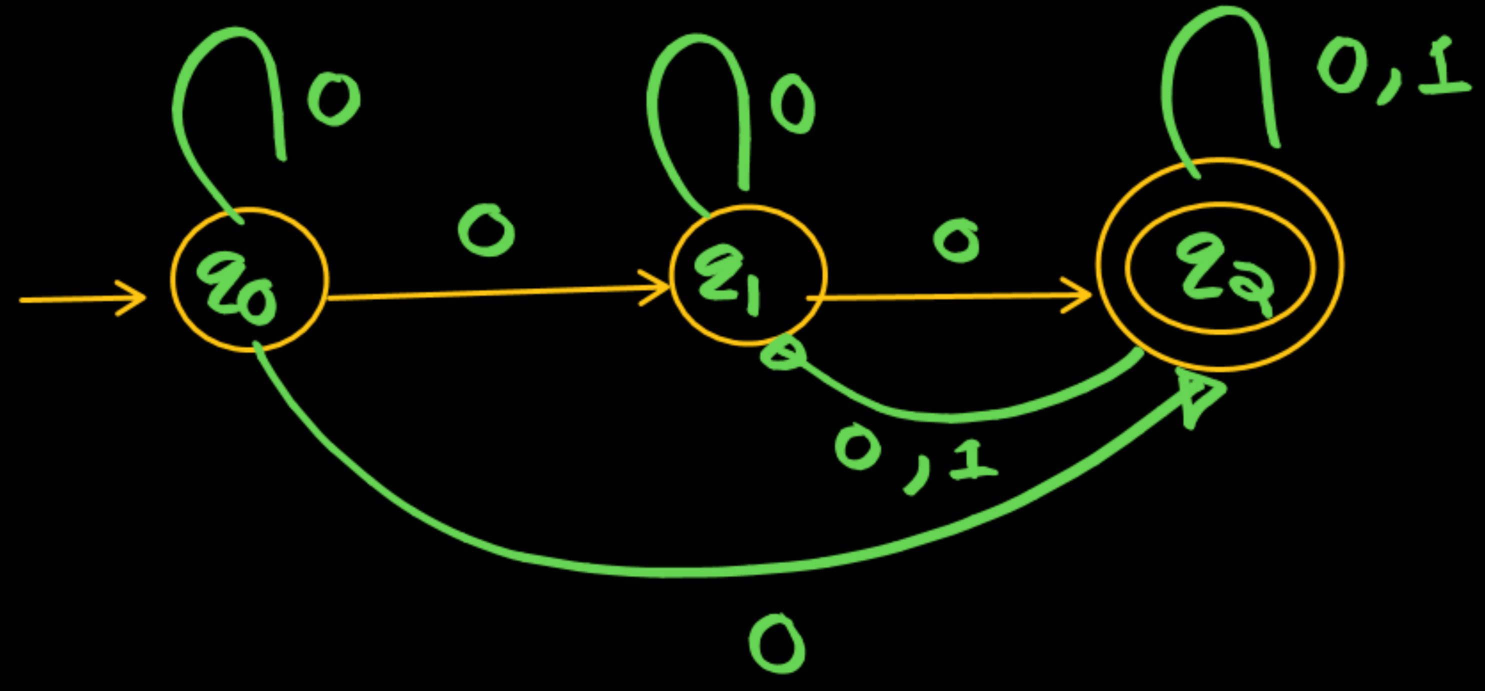
$$\epsilon\text{ closure}(q_0) = \{q_0, q_1\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1\}$$

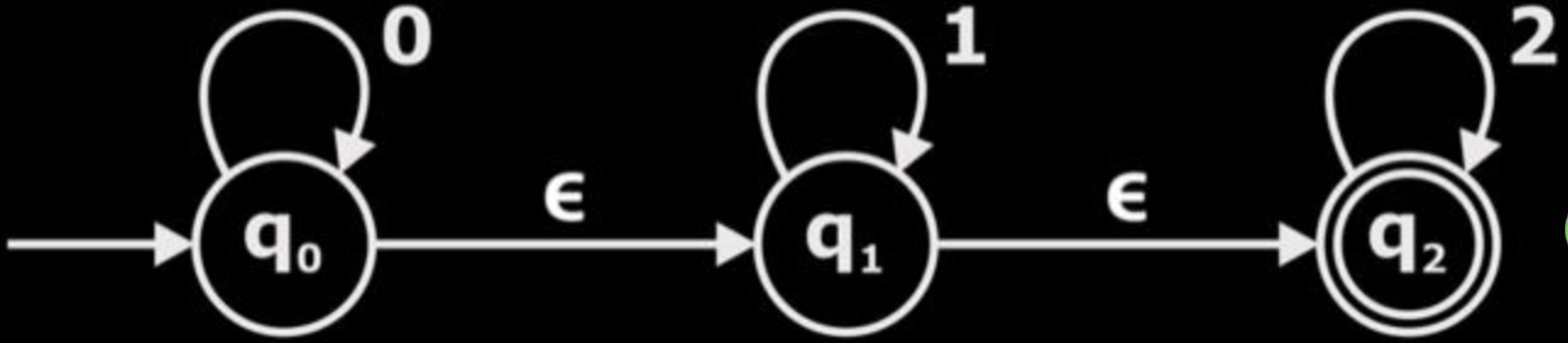
$$\epsilon\text{-closure}(q_2) = \{q_2, q_1\}$$

NFA table

δ'	0	1
q_0	q_0, q_1, q_2	ϕ
q_1	q_2, q_1	ϕ
q_2	q_2, q_1	q_2, q_1



Example-4



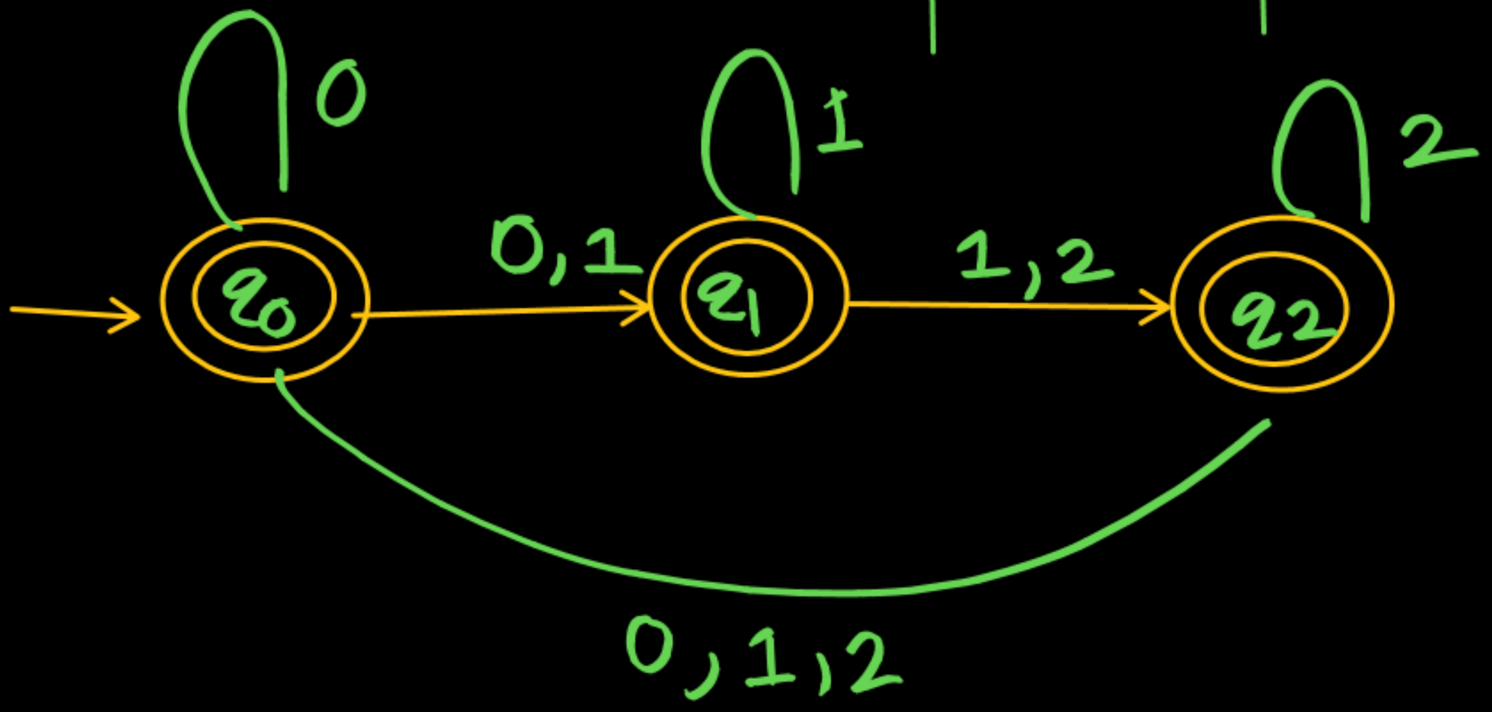
ϵ closure$(q_0) = \{q_0, q_1, q_2\}$

ϵ closure$(q_1) = \{q_1, q_2\}$

ϵ closure$(q_2) = \{q_2\}$

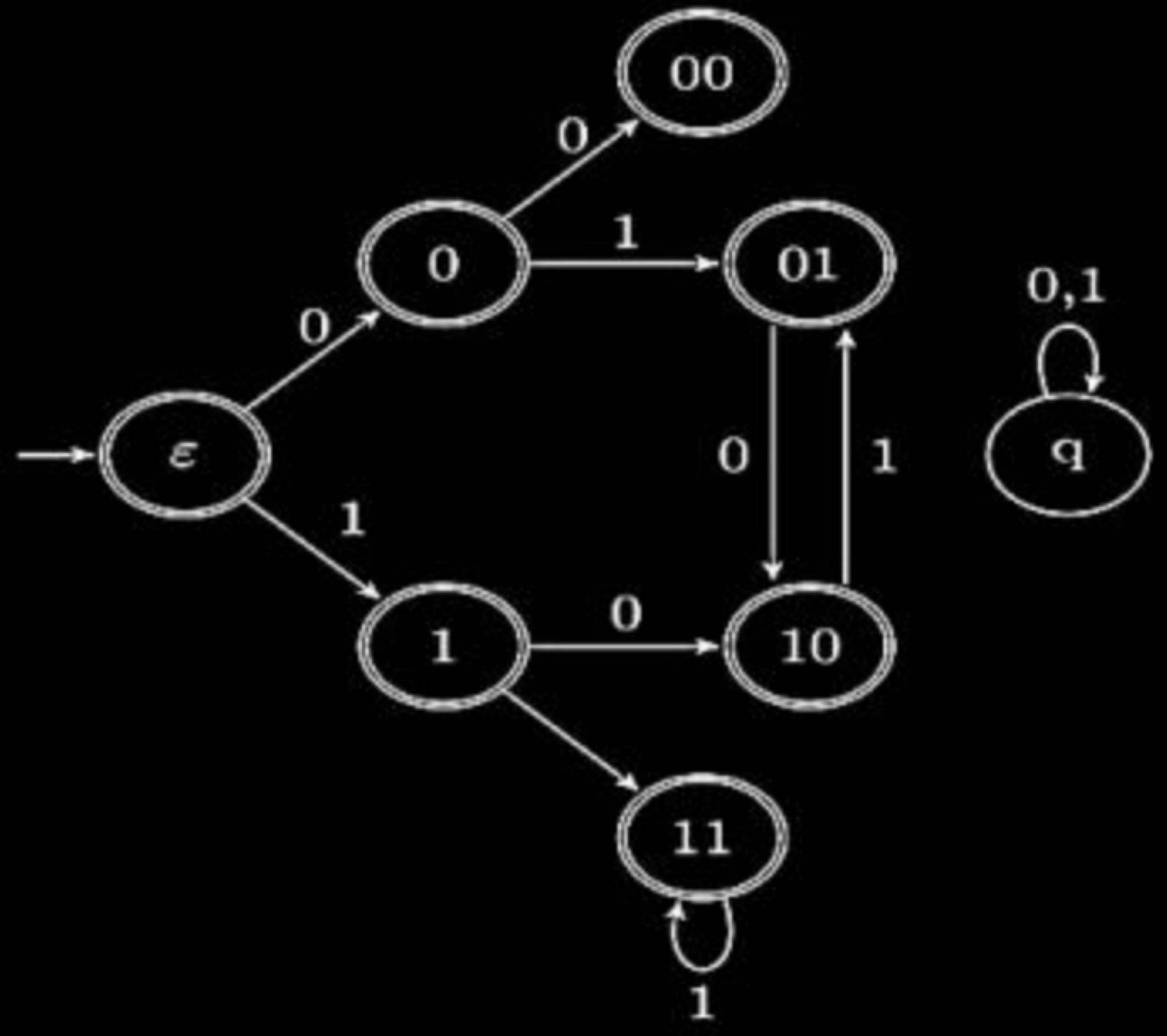
ADRule

δ'	0	1	2
q_0	q_0, q_1, q_2	q_1, q_2	q_2
q_1	ϕ	q_1, q_2	q_2
q_2	ϕ	ϕ	q_2



Question

Consider the set of strings on {0, 1} in which, every substring of 3 symbols has at most two zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed DFA that accepts this language is shown below.



Question

The missing arcs in the DFA are:

A

	00	01	10	11	q
00	1	0			
01				1	
10	0				
11			0		

B

	00	01	10	11	q
00		0			1
01		1			
10				0	
11		0			

C

	00	01	10	11	q
00		1			0
01		1			
10			0		
11		0			

D

	00	01	10	11	q
00		1			0
01				1	
10	0				
11			0		

Next class → 28^{th} November

Thank You

21^{st} Nov → AI → Sridhar sir

$\boxed{21-27}$ → AI Completed → TOC